

Cauchy Problems with Time-Dependent One-Sided Lipschitz Vector Fields

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Abstract

We study the well-posedness of Cauchy problems for ordinary differential equations whose vector fields are continuous and satisfy a *time-dependent one-sided Lipschitz condition* with integrable Lipschitz constant. This framework is natural in diffusion models, and is strictly weaker than classical global Lipschitz assumptions.

1 The Cauchy Problem

1.1 Setting

Let $v : [0, \infty) \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ and consider the Cauchy problem

$$\begin{cases} \dot{x}(t) = v_t(x(t)), & t > 0, \\ x(0) = x_0 \in \mathbb{R}^d. \end{cases} \quad (1)$$

1.2 Assumptions

We make the following assumptions.

(H1) **Continuity.** The map $(t, x) \mapsto v_t(x)$ is continuous on $[0, \infty) \times \mathbb{R}^d$.

(H2) **Time-dependent one-sided Lipschitz condition.** There exists a measurable function $L : [0, \infty) \rightarrow \mathbb{R}$ such that

$$\langle v_t(x) - v_t(y), x - y \rangle \leq L(t) \|x - y\|^2 \quad \forall t \geq 0, \forall x, y \in \mathbb{R}^d, \quad (2)$$

with, for all $T > 0$,

$$\int_0^T L(t) dt < \infty.$$

(H3) **One-sided growth condition.** There exist functions $a, b \in L^1_{\text{loc}}([0, \infty))$ with $a(t), b(t) \geq 0$ such that

$$\langle v_t(x), x \rangle \leq a(t) \|x\|^2 + b(t) \quad \forall t \geq 0, \forall x \in \mathbb{R}^d. \quad (3)$$

2 Local Existence

Theorem 2.1 (Peano). *Under assumption (H1), the Cauchy problem (1) admits at least one local solution*

$$x \in C^1([0, T]; \mathbb{R}^d)$$

for some $T > 0$.

Proof. This is a direct application of Peano's existence theorem, which requires only continuity of the vector field. $\square$

3 Uniqueness

Lemma 3.1 (Grönwall estimate under time-dependent OSL). *Let x_1, x_2 be two solutions of (1) defined on $[0, T]$. Then for all $t \in [0, T]$,*

$$\|x_1(t) - x_2(t)\| \leq e^{\int_0^t L(s) ds} \|x_1(0) - x_2(0)\|.$$

Proof. Define $\phi(t) = \|x_1(t) - x_2(t)\|^2$. Differentiating,

$$\phi'(t) = 2\langle v_t(x_1(t)) - v_t(x_2(t)), x_1(t) - x_2(t) \rangle.$$

Using (2),

$$\phi'(t) \leq 2L(t)\phi(t).$$

Applying Grönwall's inequality yields the claim. $\square$

Theorem 3.2 (Uniqueness). *Under assumptions (H1)–(H2), the Cauchy problem (1) admits at most one solution.*

Proof. If $x_1(0) = x_2(0)$, Lemma 3.1 implies $x_1(t) = x_2(t)$ for all t . $\square$

4 Global Existence

4.1 A Priori Norm Estimate

Lemma 4.1 (Norm bound). *Let $x(t)$ be a solution of (1). Then for all $t \geq 0$,*

$$\|x(t)\|^2 \leq e^{2\int_0^t a(s) ds} \left(\|x_0\|^2 + \int_0^t 2b(s)e^{-2\int_0^s a(r) dr} ds \right).$$

Proof. Set $E(t) = \|x(t)\|^2$. By differentiability,

$$E'(t) = 2\langle v_t(x(t)), x(t) \rangle.$$

Using (3),

$$E'(t) \leq 2a(t)E(t) + 2b(t).$$

The estimate follows by a standard Grönwall argument. $\square$

4.2 Continuation Argument

Theorem 4.2 (Global existence). *Under assumptions (H1)–(H3), the unique solution of the Cauchy problem (1) is defined for all $t \geq 0$.*

Proof. Let $[0, T_{\max})$ be the maximal interval of existence. Lemma 4.1 shows that $\|x(t)\|$ remains bounded on any finite interval. By the classical continuation criterion for ODEs, finite-time blow-up is therefore impossible, and $T_{\max} = \infty$. $\square$

5 Main Result

Theorem 5.1 (Global well-posedness). *Under assumptions (H1)–(H3), the Cauchy problem (1) admits a unique global solution*

$$x \in C^1([0, \infty); \mathbb{R}^d).$$

Proof. Combine Theorems 2.1, 3.2, and 4.2. $\square$

6 Numerical Euler Discretization

Fix a step size $h > 0$ and define the explicit Euler scheme on the grid $t_k := kh$:

$$x_{k+1} = x_k + h v_{t_k}(x_k), \quad x_0 \in \mathbb{R}^d. \quad (4)$$

Let $x(t)$ denote the (unique) solution of (1) and set the grid error

$$e_k := \|x(t_k) - x_k\|.$$

6.1 Stability via the Lipschitz constant of the Euler map

For each $t \geq 0$, introduce the Euler update map

$$F_t(x) := x + h v_t(x), \quad K(t) := \text{Lip}(F_t) = \sup_{x \neq y} \frac{\|F_t(x) - F_t(y)\|}{\|x - y\|}.$$

Then, for all $k \geq 0$,

$$e_{k+1} \leq K(t_k) e_k + \|\delta_{k+1}\|, \quad (5)$$

where the local truncation error is

$$\delta_{k+1} := x(t_{k+1}) - \left(x(t_k) + h v_{t_k}(x(t_k)) \right).$$

Indeed,

$$x(t_{k+1}) - x_{k+1} = \delta_{k+1} + \left(F_{t_k}(x(t_k)) - F_{t_k}(x_k) \right),$$

hence $\|F_{t_k}(x(t_k)) - F_{t_k}(x_k)\| \leq K(t_k) \|x(t_k) - x_k\| = K(t_k) e_k$, which gives (5).

Iterating (5) and using $e_0 = 0$ yields the discrete Grönwall bound

$$e_N \leq \sum_{j=0}^{N-1} \|\delta_{j+1}\| \prod_{\ell=j+1}^{N-1} K(t_\ell), \quad N \geq 1. \quad (6)$$

In particular, if $K := \sup_{t \in [0, T]} K(t) < \infty$ and $Nh \leq T$, then

$$e_N \leq \|\delta\|_{\max} \sum_{m=0}^{N-1} K^m = \begin{cases} \|\delta\|_{\max} \frac{K^N - 1}{K - 1}, & K \neq 1, \\ N \|\delta\|_{\max}, & K = 1, \end{cases} \quad (7)$$

where $\|\delta\|_{\max} := \max_{0 \leq j \leq N-1} \|\delta_{j+1}\|$.

6.2 One-sided Lipschitz refinement

Assume in addition the time-dependent one-sided Lipschitz (OSL) condition: there exists $L : [0, \infty) \rightarrow \mathbb{R}$ measurable such that, for all $t \geq 0$ and $x, y \in \mathbb{R}^d$,

$$\langle v_t(x) - v_t(y), x - y \rangle \leq L(t) \|x - y\|^2. \quad (8)$$

Introduce the squared error $E_k := e_k^2 = \|x(t_k) - x_k\|^2$. Using the decomposition

$$x(t_{k+1}) - x_{k+1} = \delta_{k+1} + (F_{t_k}(x(t_k)) - F_{t_k}(x_k)),$$

we expand

$$\begin{aligned} E_{k+1} &= \|\delta_{k+1} + F_{t_k}(x(t_k)) - F_{t_k}(x_k)\|^2 \\ &\leq (1 + \eta) \|F_{t_k}(x(t_k)) - F_{t_k}(x_k)\|^2 + \left(1 + \frac{1}{\eta}\right) \|\delta_{k+1}\|^2, \quad \eta > 0. \end{aligned} \quad (9)$$

To control the first term, note that

$$\begin{aligned} \|F_t(x) - F_t(y)\|^2 &= \|x - y + h(v_t(x) - v_t(y))\|^2 \\ &= \|x - y\|^2 + 2h \langle v_t(x) - v_t(y), x - y \rangle + h^2 \|v_t(x) - v_t(y)\|^2. \end{aligned} \quad (10)$$

Hence, under (8),

$$\|F_t(x) - F_t(y)\|^2 \leq \left(1 + 2hL(t)\right) \|x - y\|^2 + h^2 \|v_t(x) - v_t(y)\|^2. \quad (11)$$

In particular, if we also have a (possibly time-dependent) spatial Lipschitz bound

$$\|v_t(x) - v_t(y)\| \leq \Lambda(t) \|x - y\| \quad \forall t \geq 0, \forall x, y, \quad (12)$$

then combining (11) with (12) gives the refined Euler-step stability estimate

$$\|F_t(x) - F_t(y)\| \leq \tilde{K}(t) \|x - y\|, \quad \tilde{K}(t) := \sqrt{1 + 2hL(t) + h^2 \Lambda(t)^2}. \quad (13)$$

Plugging (13) into (9) yields, for all $k \geq 0$,

$$E_{k+1} \leq (1 + \eta) \tilde{K}(t_k)^2 E_k + \left(1 + \frac{1}{\eta}\right) \|\delta_{k+1}\|^2. \quad (14)$$

Iterating (14) and using $E_0 = 0$ gives a discrete Grönwall bound in energy form: for $N \geq 1$,

$$E_N \leq \left(1 + \frac{1}{\eta}\right) \sum_{j=0}^{N-1} \|\delta_{j+1}\|^2 \prod_{\ell=j+1}^{N-1} (1 + \eta) \tilde{K}(t_\ell)^2. \quad (15)$$

When $\tilde{K}(t) \leq \tilde{K}$ uniformly on $[0, T]$ and $Nh \leq T$, this simplifies to

$$e_N \leq \sqrt{\left(1 + \frac{1}{\eta}\right) \|\delta\|_{\max}^2 \sum_{m=0}^{N-1} ((1 + \eta) \tilde{K}^2)^m}. \quad (16)$$

The constant $\tilde{K}(t) = \sqrt{1 + 2hL(t) + h^2 \Lambda(t)^2}$ can be substantially smaller than the crude bound $K(t) \leq 1 + h\Lambda(t)$, especially in dissipative regimes where $L(t) < 0$.

6.3 A first-order global error bound

Assume that v is differentiable and define the acceleration bound

$$A_{\max} := \left\| \partial_t v_t(x) + (D_x v_t(x)) v_t(x) \right\|_{L^\infty([0,T] \times \mathbb{R}^d)}. \quad (17)$$

Along any solution $x(t)$ of (1),

$$\frac{d}{dt} v_t(x(t)) = \partial_t v_t(x(t)) + (D_x v_t(x(t))) v_t(x(t)),$$

so (17) implies $\left\| \frac{d}{dt} v_t(x(t)) \right\| \leq A_{\max}$. A standard integral remainder argument gives the local estimate

$$\|\delta_{k+1}\| \leq \frac{1}{2} A_{\max} h^2, \quad \text{whenever } t_{k+1} \leq T. \quad (18)$$

Global bound (Lipschitz form). Combining (18) with (7) yields, for $Nh \leq T$,

$$e_N \leq \frac{1}{2} A_{\max} h^2 \begin{cases} \frac{K^N - 1}{K - 1}, & K \neq 1, \\ N, & K = 1. \end{cases} \quad (19)$$

Global bound (OSL-refined energy form). Combining (18) with (16) yields, for $Nh \leq T$,

$$e_N \leq \frac{1}{2} A_{\max} h^2 \sqrt{\left(1 + \frac{1}{\eta}\right) \sum_{m=0}^{N-1} ((1 + \eta) \tilde{K}^2)^m}, \quad \tilde{K} := \sup_{t \in [0, T]} \sqrt{1 + 2hL(t) + h^2 \Lambda(t)^2}. \quad (20)$$

Remark 6.1. The OSL condition (8) does not by itself control $\|F_t(x) - F_t(y)\|$, because the expansion (10) contains the additional term $h^2 \|v_t(x) - v_t(y)\|^2$. However, when combined with a (possibly local) spatial Lipschitz control (12), it yields the refined stability factor $\tilde{K}(t) = \sqrt{1 + 2hL(t) + h^2 \Lambda(t)^2}$, which can be much smaller than the crude estimate $1 + h\Lambda(t)$ in contractive/monotone regimes, and leads to the sharper global bounds (14)–(20).